CM0845 Logic
Propositional Logic: Semantic Tableaux

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Preliminaries

Textbook

Mordechai Ben-Ari (2012). Mathematical Logic for Computer Science. Section 2.6.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Semantics Tableaux

Theorem 2.67

Let φ be a proposition and $\mathfrak{T}$ be a completed tableau for φ. The proposition φ is unsatisfiable if and only if $\mathfrak{T}$ is closed.

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Example (Exercise 2.9)

Prove that $\models (\varphi \wedge \psi \to \sigma) \to ((\varphi \to \sigma) \vee (\psi \to \sigma))$ using semantic tableaux.

References

Mordechai Ben-Ari [1993] (2012). Mathematical Logic for Computer Science. 3rd ed. Springer (cit. on p. 2).